



Pharm*Achieve*

MENOPAUSE

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AR	Absolute Risk
C/I	Contraindication
CHC	Combined Hormonal Contraceptive
CVD	Cardiovascular Disease
HRT	Hormonal Replacement Therapy
HS	At bedtime
HT	Hormonal Therapy
IUD	Intrauterine Device
IUS	Intrauterine System
LNG-IUS	Levonorgestrel-Releasing Intrauterine System
RCT	Randomized Controlled Trial
RR	Relative Risk
SERM	Selective Estrogen Receptor Modulator
TSEC	Tissue-Selective Estrogen Complex
UTI	Urinary Tract Infection
VMS	Vasomotor Symptoms

INTRODUCTION

Perimenopause: “menopausal transition” where there is an alteration in menstrual cycle length prior to final menstrual period

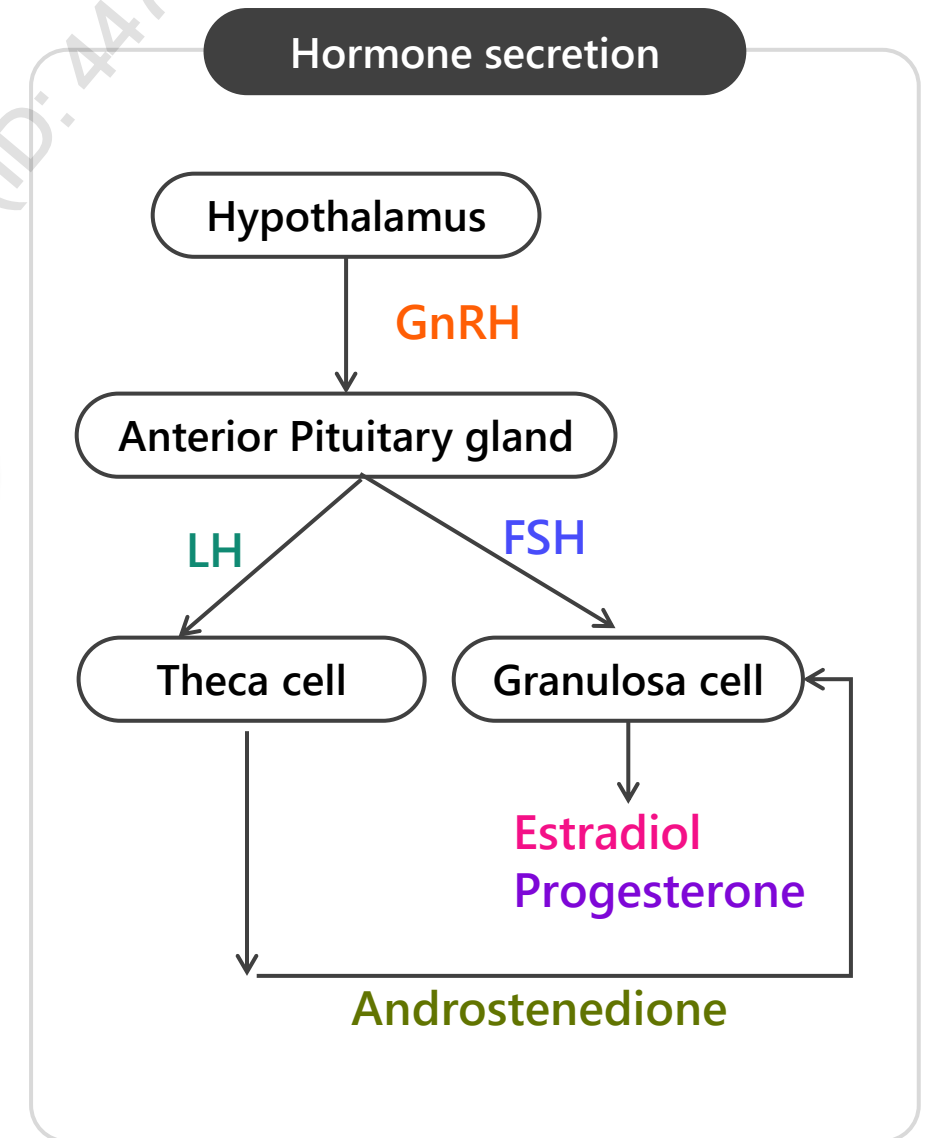
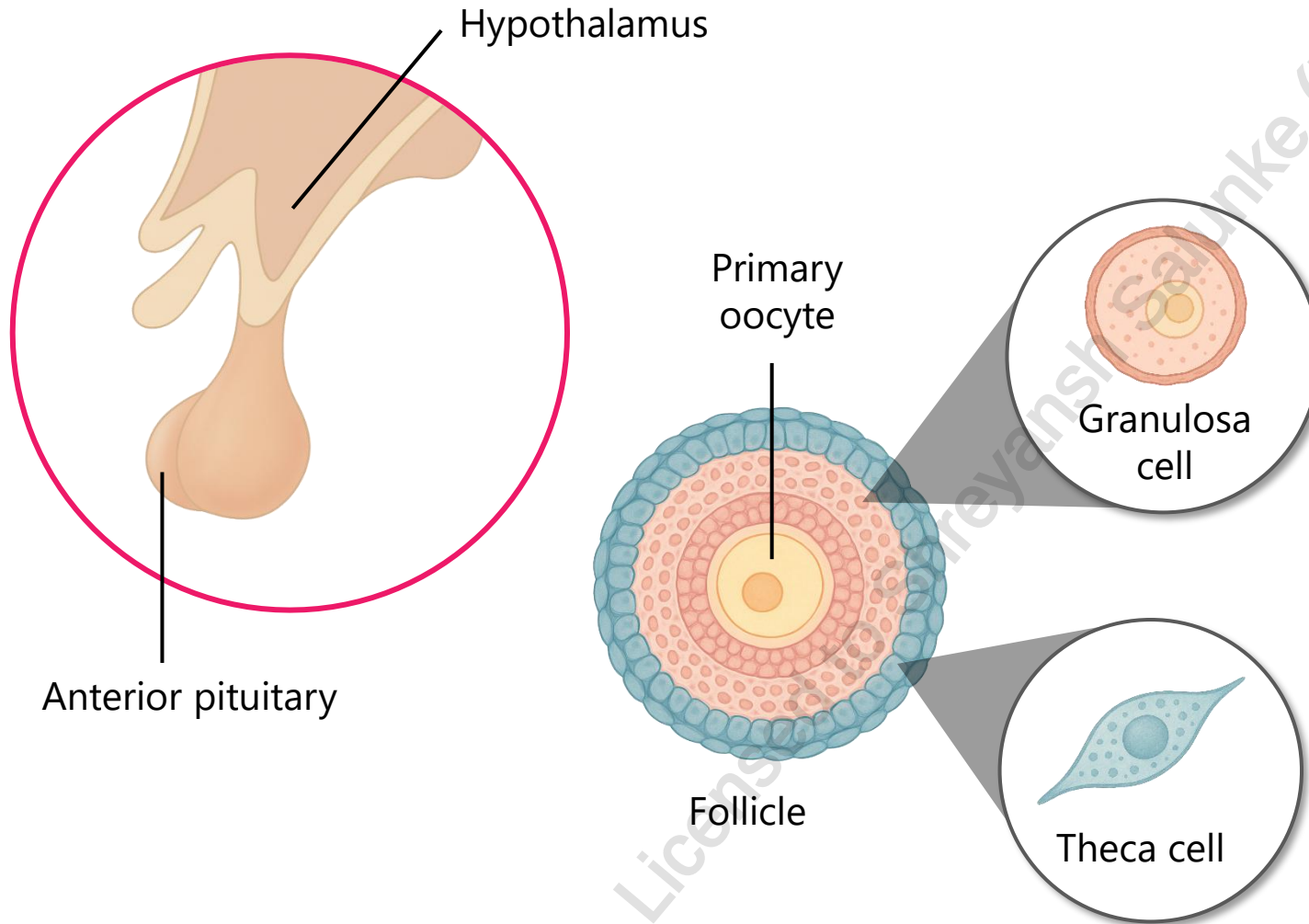
Menopause: cessation of menses for 12 months marking the end of reproductive life

- Average age: 51

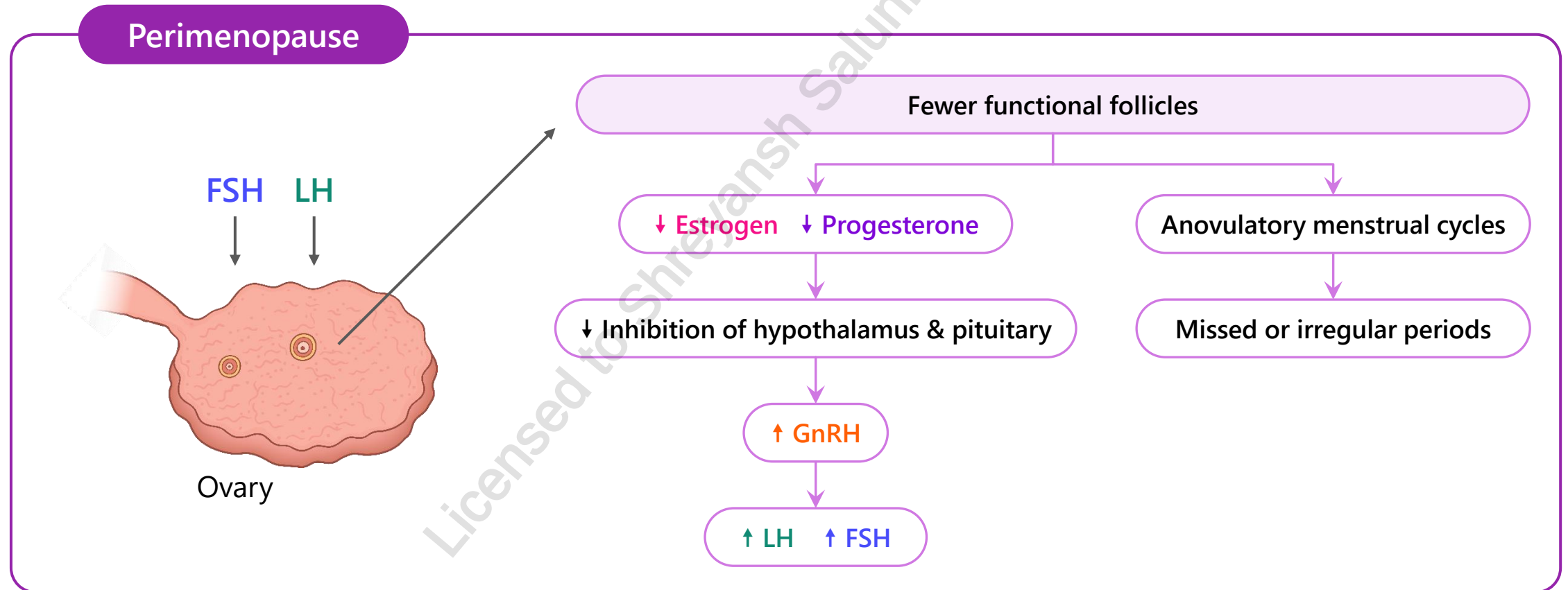
Perimenopause
(average 4 years)

Menopause

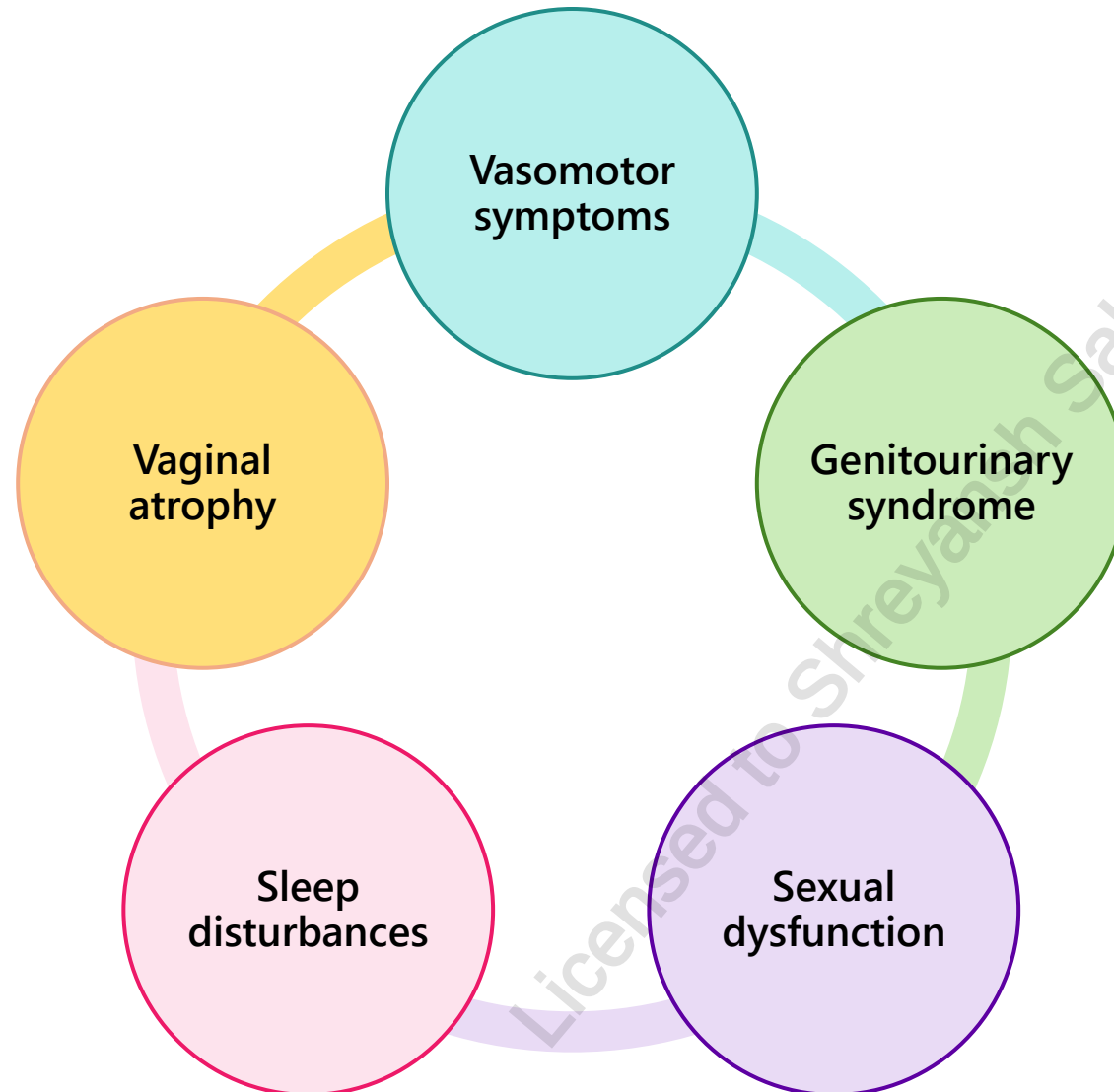
PHYSIOLOGY



PATHOPHYSIOLOGY

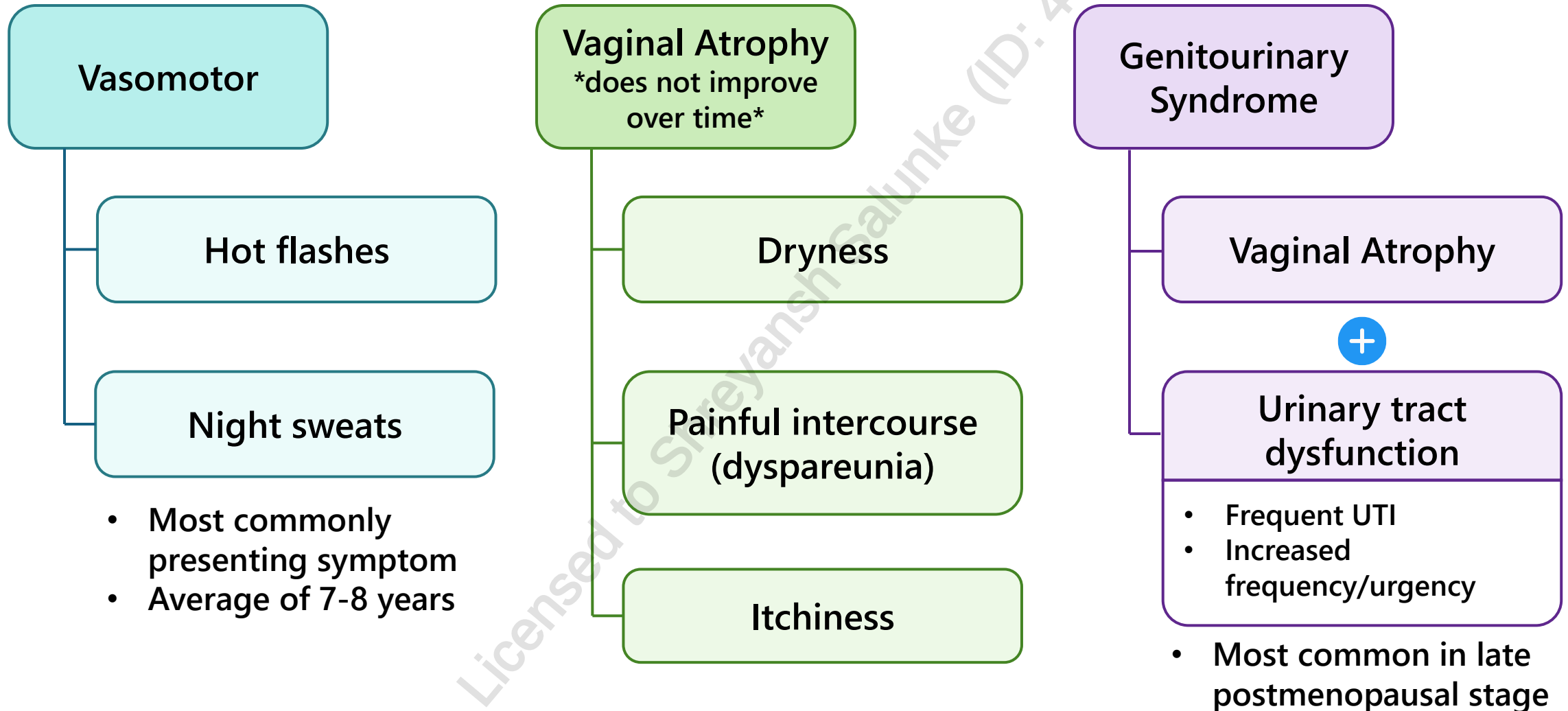


CLINICAL PRESENTATION



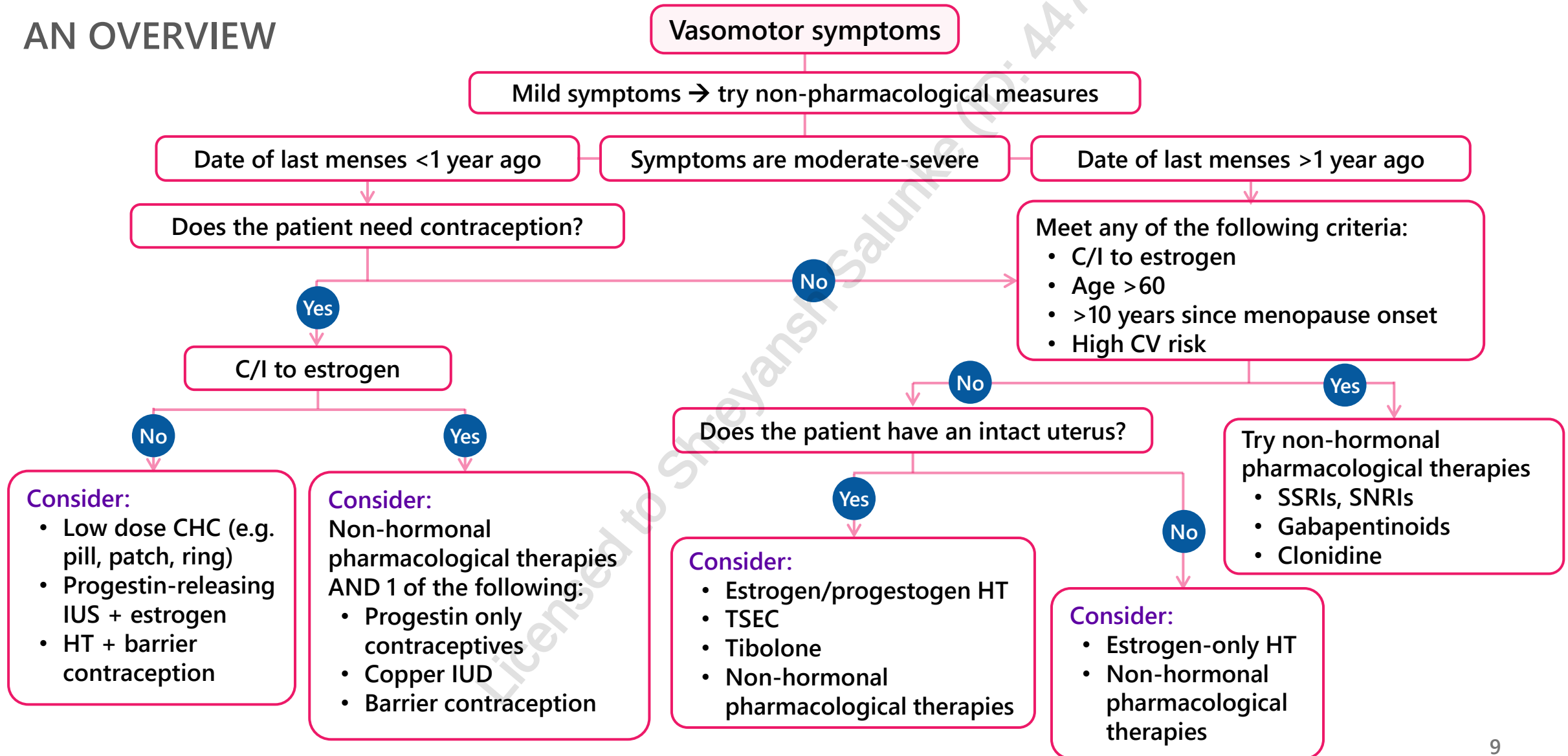
**Long-term complications:
increased risk of
osteoporosis, CVD, cancers**

CLINICAL PRESENTATION



TREATMENT ALGORITHM

AN OVERVIEW



NON-PHARMACOLOGICAL MEASURES

- Cooling techniques: **lack of evidence but may be beneficial**
- Avoiding triggers (e.g. caffeine, spicy foods): **lack of evidence but may be beneficial**
- Cognitive Behavioral Therapy (CBT): **shown efficacy in some trials**
- Weight loss: **helps reduce vasomotor symptoms**
- Regular exercise and yoga: **shown to improve symptoms**
- Vaginal lubricants and moisturizers: **shown to improve symptoms. Counselling point: only water-based lubricants are safe to use with condoms**
 - Vaginal moisturizers when used 2x per week improve dyspareunia + dryness
 - Example: Replens (offers more long-term relief)
 - **Lubricants can be used PRN basis**
 - Example: K-Y Jelly® (offers short-term rapid relief)
- Smoking cessation: smoking increases vasomotor symptom frequency and severity

PHARMACOLOGICAL MEASURES

VASOMOTOR SYMPTOMS

	Hormone Therapy	Combined Estrogen and SERM Modulator
Examples	• Progestins e.g. medroxyprogesterone acetate • Estrogens e.g. estradiol	• Estrogen-bazedoxifene
Indication	• Vasomotor symptoms in women with <u>intact uterus</u>: estrogen + progesterone • Vasomotor symptoms in women with <u>hysterectomy</u>: estrogen • <60 years of age or <10 years past menopause	• Women with uterus wishing to avoid progestogens • Moderate-severe vasomotor symptoms in women < 75 years

PHARMACOLOGICAL MEASURES

VASOMOTOR SYMPTOMS (CONTINUED)

	Hormone therapy	Combined estrogen and SERM modulator
Adverse Effects	• Estrogen: headache, dizziness, abdominal pain, rash, breast tenderness • Progesterone: breakthrough bleeding, breast tenderness, dizziness, headache, abdominal pain	• Dizziness, muscle spasm, diarrhea, nausea, dyspepsia, abdominal pain, oropharyngeal pain
Comments	• Administration: PO, transdermal (gel, patch), vaginal, IUD • The LNG-IUS, Mirena® may be used off-label (i.e. not Health Canada approved for menopause) → Mirena® is the only LNG-IUS with evidence for endometrial protection from estrogen (LNG-IUS + estrogen) • Non-oral route reduces risks of adverse effects compared to PO formulations e.g. elevated triglycerides, risk of VTE • To manage dizziness with progesterone, take at bedtime. Or split the total daily dose and take larger dose at bedtime.	• Administered PO • Avoids AEs of progestogens e.g., breast tenderness, uterine bleeding • Reduces bone resorption (may prevent osteoporosis but not yet indicated)

PHARMACOLOGICAL MEASURES

VASOMOTOR SYMPTOMS

Contraindications to Estrogen

- Undiagnosed abnormal vaginal bleeding
- Known, suspected, or history of breast cancer
- Known or suspected estrogen-dependent cancers (i.e., endometrial, ovarian)
- Coronary heart disease
- Active or history of venous thromboembolism
- Active or history of stroke
- Known thrombophilia
- Active liver disease
- Known or suspected pregnancy

Contraindications to Progestogen

- Undiagnosed abnormal vaginal bleeding
 - Current or history of breast cancer
 - Allergy to soy and peanut*
- * Prometrium® contains sunflower oil instead of peanut oil. However, due to possible cross-reactivity between soy and peanut, a peanut allergy warrants avoidance of progesterone products (brand and generic)

Contraindications to Combined Estrogen and SERM Modulator

- Patients with history of active thromboembolic disease
- Known/suspected carcinomas
- Undiagnosed genital bleeding

RISKS ASSOCIATED WITH HORMONAL THERAPIES

CV risks	• Estrogen may be cardioprotective if initiated at onset of menopause but harmful if started >10 years or if >60 years
Thromboembolic risks	• VTE risk was significantly greater in estrogen-progestin therapy • Small increase in risk of ischemic stroke • Patients with a moderate risk for CVD: may want to consider transdermal or low-estrogen dose
Breast cancer	• Incidence was higher in patients using estrogen-progesterone therapy** • Avoid HT in a patient with a history of breast cancer • Consider non-hormonal alternatives
Other risks	• Significantly elevated risk of acute pancreatitis, gall bladder disease, nephrolithiasis • HRT can worsen microvascular complications in patients with diabetes

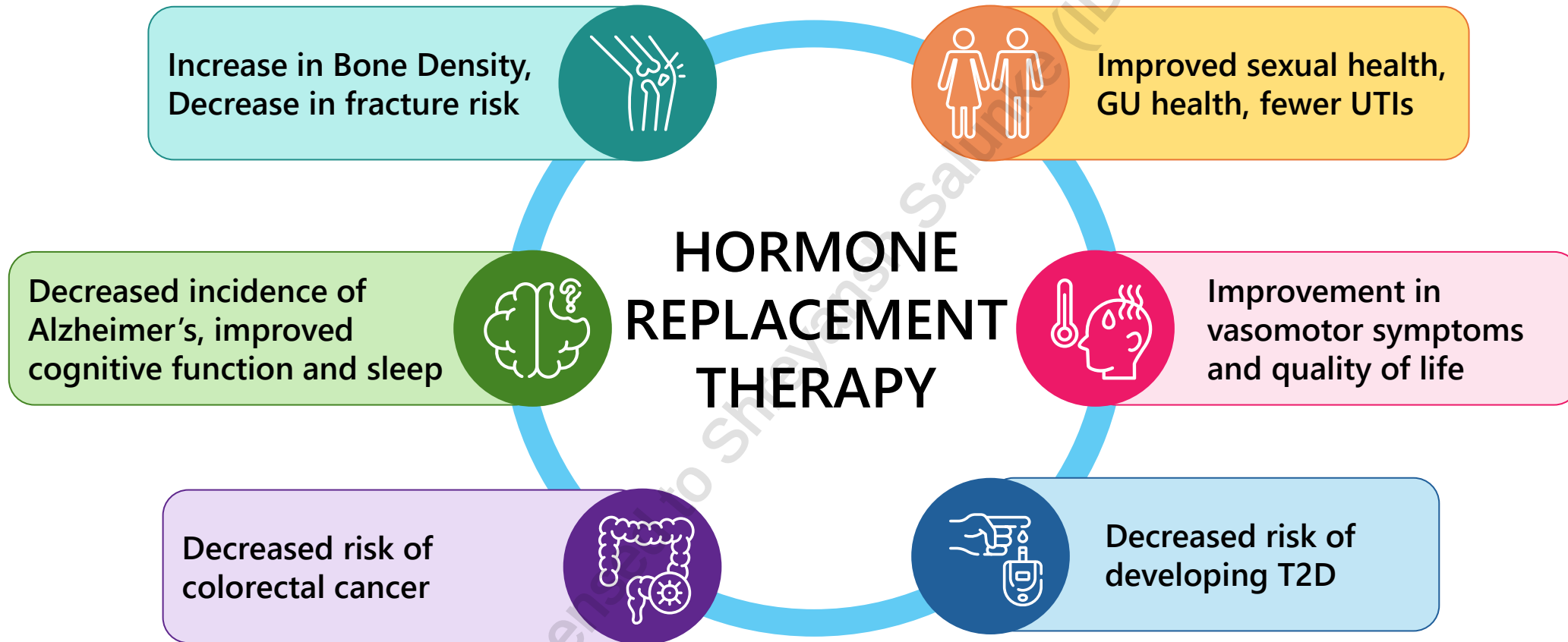


Limit HRT duration to 4-5 years



**RR: +26% vs AR: 0.08% (8/10000 pts) -FYI

BENEFITS ASSOCIATED WITH HORMONAL THERAPIES



EXAMPLES OF PRODUCTS

ORAL ESTROGENS

Name	Dosage	Comments
<i>Conjugated estrogens</i> Premarin®	Recommended starting dosage: 0.3-0.625 mg/day	- Administer with a progestogen <i>in</i> patients with an intact uterus (to prevent endometrial hyperplasia) - Estrogen is taken continuously every day - Begin with a low/standard dose and tailor appropriately

TRANSDERMAL ESTROGENS AND ESTROGEN/PROGESTIN COMBINATIONS

Name	Dosage	Comments
<i>17-beta estradiol patch</i> Estradot®	Recommended starting dose: 25–50 mcg 2x/week	Avoids 1st pass via liver. It could be beneficial in individuals who: smoke, malabsorption syndrome, elevated triglycerides, shift workers
<i>17-beta estradiol/ norethindrone acetate</i> Estalis®	Recommended starting dose: 1 patch applied twice weekly	

EXAMPLES OF PRODUCTS

ORAL ESTROGEN + PROGESTIN COMBINATIONS

Name	Dosage	Comments
<i>17-beta estradiol/ drospirenone</i> Anelq®	Recommended dose: 1 tablet daily PO continuously (1 mg/1 mg)	With continuous use patients will experience reduced menstrual flow and amenorrhea

ORAL PROGESTOGENS

Name	Dosage	Comments
<i>Medroxyprogesterone acetate</i>	Recommended starting dose: Cyclic regimen: 5 mg daily for 12-14 days every month Continuous regimen: 2.5 mg/daily	- Can be sedating - To be combined with estrogen in patients with intact uterus (higher progestogen needed with higher estrogen doses) - Continuous regimen will prevent menstrual flow. However, during the first year, spotting or vaginal bleeding may still occur - Cyclical: patient will have a withdrawal bleed when progestogen is stopped. - Continuous therapy may be more effective at prevention of hyperplasia
<i>Progesterone, micronized</i> - Generic- peanut oil, soy - Prometrium® - sunflower oil, soy	Recommended starting dose: Cyclic regimen: 200 mg QHS for 12–14 days every month Continuous: 100 mg QHS	

EXAMPLES OF PRODUCTS

SYNTHETIC STEROID ANALOGUE OF NORETHYNODREL		
Name	Dosage	Comments
<i>Tibolone</i> Tibella®	Recommended dose: 2.5 mg PO daily continuously	• Take continuously for treatment of VMS at lowest effective dose for shortest duration • Concomitant progestogen not needed • Only for women with intact uterus

PHARMACOLOGICAL THERAPY

NON-HORMONAL AGENTS FOR VASOMOTOR SYMPTOMS

- Citalopram, desvenlafaxine, escitalopram, paroxetine, venlafaxine
 - Consider if HT is contraindicated or undesirable
 - Provides mild-moderate reduction in frequency and severity of hot flashes
- Gabapentin
 - Reduced frequency & severity of hot flashes in a small trial in women experiencing ≥ 7 hot flashes/day
 - ADR: drowsiness → but it could be beneficial as a HS dosing for patients who have night sweats and sleep disturbances.
- Clonidine
 - May be used if benefit > risk of adverse effects
- Herbal products: black cohosh, omega-3 fatty acids, among others
 - Insufficient or inconclusive data
- Vitamin E
 - Marginal benefit on symptoms
- Soy isoflavones
 - Inconclusive evidence - may improve hot flashes and vaginal dryness, but not night sweats



Black cohosh CI
in liver disease

PHARMACOLOGICAL THERAPY

VAGINAL SYMPTOMS (INTRAVAGINAL ESTROGEN)

Available as creams, tablets or rings

Indicated in vaginal atrophy (localized symptoms)

Shown to decrease UTI recurrence & improve urinary incontinence

Safety beyond 1 year has not been established in RCT's



Question: When using a recommended dose of intravaginal estrogen, is a progestogen indicated?

EXAMPLES OF PRODUCTS

VAGINAL ESTROGEN		
Name	Dosage	Comments
<i>Conjugated estrogen vaginal cream</i> Premarin Vaginal Cream	0.5 g daily x 14 days, followed by 0.5 g 2–3 times weekly	- All forms are equally effective - ADR: Local burning, vaginal leakage and some irritation - Vaginal estrogen therapy displayed no increase in endometrial hyperplasia. Thus, progestogen therapy is not required, if ≤ 1 year
<i>17-beta estradiol vaginal ring</i> Estring	1 ring to be inserted Q3 months vaginally	
<i>17-beta estradiol vaginal tablet</i> Vagifem 10	Insert 10 mcg intravaginal tablet once daily x 14 days then twice weekly	

* Imvexxy® is another example of vaginal estradiol, available as 4 mcg and 10 mcg vaginal inserts

TAKEAWAY

- **Hormone therapy is the most effective therapy for hot flashes**
- Vasomotor symptoms last 7-8 years on average but vaginal symptoms do not improve with time
- Combined estrogen and progestogen OR estrogen agonist/antagonist if intact uterus, **estrogen alone if hysterectomy** to reduce risk of endometrial cancer
- Intravaginal estrogen for women with vaginal symptoms only. Do not need progestogen if only using for ≤ 1 year.
- Transdermal HRT preferred if patient has risk factors for thromboembolism or hypertriglyceridemia
- **Risks of HRT:** cardiovascular risks (higher risk when initiated further away from menopause), thromboembolic risks, acute pancreatitis, gallbladder disease, nephrolithiasis
- HRT duration should be limited to 4-5 years
- Some reduction in vasomotor symptoms with some SSRIs and SNRIs, gabapentin and clonidine
- Advise patient to see doctor regarding cardiovascular health risk screening, mammograms, bone mineral density test, etc.
- CHC or LNG-IUS are to be considered in the perimenopausal period (if contraception is requested). Avoid in the postmenopausal period as ethinyl estradiol is much higher than estrogen used in HT

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Vasomotor symptoms (e.g. hot flashes, night sweats)	Improved/resolved	3-4 weeks
Genitourinary symptoms (e.g. dryness, itchiness, dyspareunia, frequency/urgency)		Weeks-months

MONITORING PARAMETERS

SAFETY ENDPOINTS		
Parameter	Goal	Timeframe
AEs of medications (e.g. bloating, transient bleeding, headaches, GI disturbances, dizziness)	Minimize/prevent	While on therapy
Risks of HRT (e.g. breast cancer, CVD, thromboembolism, endometrial hyperplasia, pancreatitis, gall bladder disease)	Prevent	Routine follow-up and screening recommended as per local guidelines

FREQUENTLY ASKED QUESTIONS

1. Should a woman stop estrogen when she turns 60 years of age if she has been taking it prior?

- From the SOGC guidelines: There is no specific time frame or duration for systemic MHT (Menopausal Hormone Treatment) use. These treatments should be individualized, based on ongoing symptoms, benefits, and personal risks. Periodic re-evaluation of a patient's MHT prescription is recommended. MHT may be continued beyond the age of 65 years in some women with persistent, bothersome menopausal symptoms.
- Reference: J Obstet Gynaecol Can 2021;43(10):1188–1204 (Guideline No. 422a: Menopause: Vasomotor Symptoms, prescription Therapeutic Agents, Complementary and Alternative Medicine, Nutrition, and Lifestyle)

2. Estrogen products are contraindicated in smokers over the age of 35 in general, or it has to be more than 15 cigarettes per day as per the slides?

- Smoking is not in the list of contraindications to systemic menopausal hormone therapy (see below), it is important to conduct a risk assessment for these patients.
- From SOGC guidelines: Estrogen is available in the form of oral pills, transdermal patches and gels, and vaginal applications. The estrogens found in Canadian products include conjugated estrogens, estradiol, and estrone. Transdermal estrogen products do not have a first-pass effect through the liver and provide more consistent estrogen levels. They may be preferred over oral pills in women who are smokers or shift workers, or who have high triglyceride levels, hypertension, gall bladder disease, migraines, or malabsorption syndromes. Based on observational data alone, standard doses of transdermal estrogen may be associated with lower risk of venous thromboembolism than oral estrogen.
- Reference: J Obstet Gynaecol Can 2021;43(10):1188–1204 (Guideline No. 422a: Menopause: Vasomotor Symptoms, prescription Therapeutic Agents, Complementary and Alternative Medicine, Nutrition, and Lifestyle)
- **Contraindications to systemic menopausal hormone therapy:**

Contraindications to estrogen

- Undiagnosed abnormal vaginal bleeding
- Known, suspected, or history of breast cancer
- Known or suspected estrogen-dependent cancers (i.e., endometrial, ovarian)
- Coronary heart disease
- Active or history of venous thromboembolism
- Active or history of stroke
- Known thrombophilia
- Active liver disease
- Known or suspected pregnancy

Contraindications to progestogen

- Undiagnosed abnormal vaginal bleeding
- Current or history of breast cancer

RESOURCES

Resources that can be used in practice:



CASE

SD is a 51-year-old female who is complaining of hot flashes. SD's last menstrual cycle was 5 months ago, and she has been experiencing spotting. The hot flashes started a few weeks ago and so far, she is managing well but would like to know if there's anything she can do to minimize them. She is sexually active but does not use any protection. She doesn't want any more children. She took Plan B® a few years ago when she had a pregnancy scare and experienced a lot of unpleasant side effects and "never wants to take it again." She has osteoarthritis and orthostatic hypotension. She takes Tylenol 500 mg Q6H PRN and Voltaren® PRN for her osteoarthritis. SD is married with 2 children and her husband had a vasectomy a few years ago. She does not smoke or drink and she eats a well-balanced diet.

1. Which of the following is a correct statement?
 - a) SD is perimenopausal
 - b) SD is menopausal
 - c) SD is experiencing menstrual irregularities due to her previous Plan B® use
 - d) None of the above



CASE

SD is a 51-year-old female who is complaining of hot flashes. SD's last menstrual cycle was 5 months ago, and she has been experiencing spotting. The hot flashes started a few weeks ago and so far, she is managing well but would like to know if there's anything she can do to minimize them. She is sexually active but does not use any protection. She doesn't want any more children. She took Plan B® a few years ago when she had a pregnancy scare and experienced a lot of unpleasant side effects and "never wants to take it again." She has osteoarthritis and orthostatic hypotension. She takes Tylenol 500 mg Q6H PRN and Voltaren® PRN for her osteoarthritis. SD is married with 2 children and her husband had a vasectomy a few years ago. She does not smoke or drink and she eats a well-balanced diet.

2. Which of the following is an appropriate recommendation for SD at this point?

- a) Initiate hormonal therapy
- b) Initiate nonpharmacologic measures
- c) Counsel on contraceptive use
- d) B and C



CASE

SD is a 51-year-old female who is complaining of hot flashes. SD's last menstrual cycle was 5 months ago, and she has been experiencing spotting. The hot flashes started a few weeks ago and so far, she is managing well but would like to know if there's anything she can do to minimize them. She is sexually active but does not use any protection. She doesn't want any more children. She took Plan B® a few years ago when she had a pregnancy scare and experienced a lot of unpleasant side effects and "never wants to take it again." She has osteoarthritis and orthostatic hypotension. She takes Tylenol 500 mg Q6H PRN and Voltaren® PRN for her osteoarthritis. SD is married with 2 children and her husband had a vasectomy a few years ago. She does not smoke or drink and she eats a well-balanced diet.

3. SD returns to you after a few weeks as her symptoms are worsening.
What is the appropriate next step?

- a) Initiate combined HT
- b) Initiate estrogen-only therapy
- c) Initiate estrogen-basedoxifene
- d) Initiate clonidine



CASE

SD is a 51-year-old female who is complaining of hot flashes. SD's last menstrual cycle was 5 months ago, and she has been experiencing spotting. The hot flashes started a few weeks ago and so far, she is managing well but would like to know if there's anything she can do to minimize them. She is sexually active but does not use any protection. She doesn't want any more children. She took Plan B® a few years ago when she had a pregnancy scare and experienced a lot of unpleasant side effects and "never wants to take it again." She has osteoarthritis and orthostatic hypotension. She takes Tylenol 500 mg Q6H PRN and Voltaren® PRN for her osteoarthritis. SD is married with 2 children and her husband had a vasectomy a few years ago. She does not smoke or drink and she eats a well-balanced diet.

4. What is an important counselling point for SD when dispensing this new medication?
- a) Reassessment is needed every year as the therapy is life-long
 - b) If symptoms resolve, the medication can be gradually stopped
 - c) Contraceptive use is required until menopause
 - d) B and C



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CHANGE LOG

August 2025

- Formatting changes made throughout
- Slide 3: Added legend
- Slide 15: Minor modifications to tibolone
- Slide 16: Added 'inconclusive evidence' for soy isoflavones
- Slide 22: Updated link
- Slide 26: Added references

September 2025

- Slide 20-21: Added monitoring parameters slide

November 2025

- Slide 3: Updated legend
- Slides 5 and 6: Added pathophysiology slides
- Slide 14: Updated risk incidence for breast cancer, added context of risks
- Slide 15: New slide on benefits of HRT
- Slide 17: Added sedative effect of oral progesterones
- Slides 23 and 24: Updated monitoring parameters
- Slide 31: Added references

January 2026

- Slide 3: Updated legend
- Slide 12: Clarified role of the LNG-IUS Mirena® in menopause
- Slide 13: Removed 'prometrium only' as soy and peanut allergy also applies to some generic progesterone. Added note on cross-reactivity of soy and peanut allergy
- Slide 26: Added another resource
- Slide 31: Updated references

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 17: Removed brand name Provera® as no longer available as an oral formulation on Canadian market. Minor update to dosage regimen
- Slide 21: Added note on Imvexxy®